

Welcome to **instats**

The session will begin shortly.

Start

A 3-DAY WORKSHOP ON LLMS FOR MEASUREMENT

From a model to a measurement.

Walk-through, gold set, validation, time series. The pipeline that turns a notebook into a defensible proxy.

INSTRUCTOR

Jihye Park, University of Geneva

AUDIENCE

Political scientists, no Python required

DURATION

2 hours 30 minutes

Four movements, two breaks, one CSV.

MOVEMENT 01, 40 MIN

Walk-through

Load UNGDC. Score in batches. Save speech-level CSV. Aggregate to country-year.

MOVEMENT 02, 35 MIN +
FINE-TUNING

Validity check

Thirty-row gold set. Confusion matrix. Cohen's kappa. Pearson r. Subgroup table. Then SetFit fine-tuning when the kappa is too low.

MOVEMENT 03, 35 MIN

Time series

Plot climate emphasis from 1946 onward. Compare to the Paris Agreement.

MOVEMENT 04, 15 MIN

Q&A and checklist

The validation appendix in two columns. What to ship. What revisions usually ask for.

Two breaks of ten minutes total. The instructor demos run on paid Colab Pro for speed. Notebooks on the seminar webpage.

Two model families. Today is encoder day.

Yesterday you ran a zero-shot DeBERTa pipeline on roughly two hundred sentences from the UN General Debate Corpus. Three candidate labels. One pass. About ninety seconds on T4.

Clear-cut sentences with explicit climate vocabulary landed where you would expect. Mixed sentences and sentences with implicit framing were the misclassifications.

A notebook that classifies two hundred sentences is a demo. A measurement is something else.

WHERE DAY 2 PICKS UP

- Run the same model across the full corpus, not just two hundred sentences.
- Hand-code thirty paragraphs as the gold answer key.
- Compute the validation numbers your appendix needs.
- Aggregate to country-year, plot against an external event.

A measurement assigns numbers to cases in a defensible way.

PROPERTY 01

Validity

The number tracks the concept it claims to track. If your concept is "stance toward climate cooperation" and your number is "general positivity," you have a validity problem.

PROPERTY 02

Reliability

Re-running the pipeline on the same input gives the same output. Two human coders applied to the same paragraph reach the same answer.

PROPERTY 03

Comparability

Scores compare across countries and years without systematic drift. If the encoder is more confident on EU speeches than on small-economy speeches, you have a comparability problem.

Adcock and Collier predate LLMs by two decades. The framework still applies. Drop it in your appendix and methods reviewers will recognize it.

Each movement answers one reviewer question.

MOVEMENT	REVIEWER QUESTION	WHAT YOU SHIP
01 walk-through	How did you compute the score for every observation in your sample?	The notebook with model ID pinned, the candidate-label list, the speech-level CSV.
02 validity	How do you know the score actually measures the concept you claim?	Thirty-row gold set, confusion matrix, Cohen's kappa, Pearson r, Spearman rho. A SetFit fine-tune if the kappa is too low.
03 time series	Show me the score on a known case where I can check it against external reality.	Country-year time series with the Paris Agreement marked.
04 checklist	What do I need in your appendix to call this defensible?	The two-column appendix checklist, must-have and often-asked-for.

Each row is a slide on screen, an output cell in the notebook, and a paragraph in your future paper.

FIRST MOVEMENT, 40 MINUTES

The walk-through.

Load the corpus. Run the model in batches. Save the scores. Aggregate by year.

01

UN General Debate Corpus, 1946 to 2025.

SPECS

- **Source.** Harvard Dataverse, doi:10.7910/DVN/OTJX8Y.
- **License.** CC0. Fully open.
- **Size.** ~30 MB gzipped, 4,761 stratified speeches.
- **Coverage.** Heads of delegation, opening of the UN General Assembly.
- **For today.** Pre-built sample bundled in the workshop release.

COLUMNS WE USE

```
# speech-level dataframe
country      # ISO3 alpha
session      # UN GA session number
year         # 1946 to 2025
text         # full speech, English
```

We will not sentence-tokenize today. Working at the speech level keeps the live encoder pass under five minutes on T4.

Each step is a cell in the notebook.

01 LOAD

Read CSV

Fetch the workshop sample from the GitHub release, cache locally. Already stratified by year.

02 TRUNCATE

To 512 tokens

DeBERTa context window. Speeches average two thousand words. Document the choice.

03 SCORE

Zero-shot, batches

Three candidate labels.
`batch_size=16` on T4.
About three minutes total.

04 SAVE

Speech-level CSV

One row per speech.
Country, year, top label, top score, supportive probability.

05 AGGREGATE

Country-year CSV

Group by country and year.
Share supportive. Mean supportive probability.
Always keep n.

Load and sample.

```
# cache the 30 MB gzipped CSV
if not os.path.exists("ungdc_sample.csv.gz"):
    urllib.request.urlretrieve(URL, "ungdc_sample.csv.gz")

ungdc = pd.read_csv("ungdc_sample.csv.gz",
                    compression="gzip")

# 4,761 rows already stratified by year
```

WHAT TO POINT OUT LIVE

- **Cache the download.** Re-running the cell does not re-download.
- **Stratified by year.** The right default for a measurement project that cares about temporal trends.
- **Year-coverage histogram.** Bottom of the cell output. Coverage should look balanced across decades.

Score in batches.

```
from transformers import pipeline
zsc = pipeline(
    "zero-shot-classification",
    model="MoritzLaurer/DeBERTa-v3-"
        "base-mnli-fever-anli",
    device=0,
)

labels = [
    "supports stronger climate action",
    "skeptical of climate action",
    "does not discuss climate",
]

results = zsc(texts,
    candidate_labels=labels,
    batch_size=16)
```

THE OFF-TOPIC LABEL IS DOING REAL WORK

Without a third "does not discuss climate" label, every speech that does not mention climate would be forced into either supportive or skeptical. The proxy would be contaminated with off-topic noise.

The third label is the sink. It absorbs the speeches that should not contribute to the climate-stance signal.

About 4,761 speeches. Three minutes on T4 GPU.

Save speech-level. Aggregate to country-year.

```
# speech-level
scored = pd.DataFrame({
    "country":    sample.country,
    "year":       sample.year,
    "top_label":  [r["labels"][0]
                   for r in results],
    "p_supportive": [supp_prob(r)
                    for r in results],
})
scored.to_csv("ungdc_speech.csv")

# country-year
cy = (scored
     .groupby(["country", "year"])
     .agg(n=("top_label", "size"),
          share=("top_label", "supp_share"),
          mean_p=("p_supportive", "mean")))
```

THE DELIVERABLE

One row per country-year. Three new columns.
Merges into your regression dataset on country and year.

Share supportive is a count. Mean supportive probability uses the underlying continuous signal.

Always keep n. Country-years with one speech are not the same as country-years with thirty.

One notebook, one model ID, one prompt list.

PROPERTY 01

Pin the model

Column name encodes provenance. score_deberta_v3_base. The HF model ID is in the script. Weights shipped with the repo.

PROPERTY 02

Pin the labels

Candidate labels documented in a sidecar text file. Anyone re-running the notebook gets the same prompts.

PROPERTY 03

Pin the seed

Random sample uses a fixed seed. Two runs of the notebook produce the same scored CSV byte-for-byte.

If a reviewer asks where the column came from, the answer is one notebook, one model ID, one prompt list.

The off-the-shelf ceiling.

CAVEAT 01

Truncation

The 512-token cut drops most of each speech. For "speech mentions climate," the lead two pages are enough. For "speech proposes specific policies," they are not. Document the choice.

CAVEAT 02

Wording sensitivity

The supportive label is one wording. A different wording would give different scores. Movement 02 puts a number on the wording sensitivity through the gold set.

CAVEAT 03

Translation drift

UNGDC speeches are translated into English by the corpus maintainers. Translation introduces drift the encoder cannot quantify. Worth a paragraph in the appendix.

None of the three is a problem to solve today. All three belong in a real paper's limitations section. After the break, once the gold set quantifies the ceiling, we fine-tune past it with SetFit on those same thirty labels.

Stop